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Nathan Jacob

Senior Data Scientist

github
LinkedIn
Kaggle
Portfolio

Professional Summary

A Full-Stack Senior Data Scientist with experience building statistical learning systems, large-scale ML infrastructure, and production AI applications across logistics, trust & safety, conversational AI, and computer vision. Specialized in architecting low-latency, high-throughput ML systems with strong focus on experimentation, optimization, reliability, and measurable business impact. Proven track record deploying models at Billion-scale workloads on Kubernetes/Triton environments, spanning classical ML, deep learning, and LLM-powered systems.

Work Experience

Meesho Bangalore, India
Data Scientist II Jan 2024 – Present

- Built and deployed a large-scale ETA prediction system using quantile regression and causal feature derivations, improving MAE by 0.8 days over incumbent production models while reducing delivery breaches by 3%. Drove 0.7% NMV impact and reduced cancellations/RTOs through statistically calibrated uncertainty estimation. Optimized inference stack to sustain 4K RPS at <6ms p99 latency on a single CPU instance.
- Architected and deployed RAG-based customer support LLMs using LangChain; achieved SOTA level accuracy at 25% cost and reduced latency from 1500ms→40ms via Triton Inference Server and quantized PyTorch models.
- Built multilingual conversational AI system supporting 8+ Indic languages (ASR + TTS + NLU); scaled to 200K+ daily interactions with 99.9% reliability and sub-800ms p99 latency.

Data Scientist I Jul 2022 – Jan 2024

- Developed computer vision + NLP pipelines detecting counterfeit listings and product quality issues through custom object detection and classification models, optimized the framework saving $40K/month, improved recall to 99.5%.
- Spearheaded successful initiatives that led to an 80% decrease in non-compliant listing views by removing 42 lakh counterfeit listings and deactivating 10 lakh restricted products, reducing them to less than 0.1% of all platform views.

Data Science Intern Jan 2022 – Jul 2022

- Developed CNN + OCR QC automation reducing manual workload by 50% and operational costs by 80%.

Projects

AI Interviewer — Multi-agent interviewer using LangChain, CrewAI, and vector search for contextual dialogue flow.
EpicAI (Shopify App) — LLM-powered CRM with RAG, Docker, NeonDB, and SPLADE v3 for real-time search and recommendations.
Universal Logo Detection — YOLOv7-based open-source detector with 1K+ monthly users.
NIAT Masterclass — Speaker on "Quantisation Fundamentals: Compressing LLMs for Speed and Cost."

Skills

Languages: Python, C/C++, SQL, GraphQL
Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, CrewAI, FastAPI, Flask
MLOps & Cloud: Azure, MLflow, Airflow, Docker, Kubernetes, Databricks, CI/CD, Git
Specializations: LLM Fine-Tuning, RAG, Prompt Engineering, Multi-Agent Systems, Quantization, Generative AI, Model Optimization, Compliance-Aware AI Deployment

Certifications

Deep Learning Specialization — DeepLearning.AI, Jun 2020
Machine Learning — Stanford University, May 2020

Education

Indian Institute of Information Technology, Allahabad Prayagraj, India
B.Tech, Electronics and Communication Engineering Aug 2018 – Jul 2022

Awards

- Kaggle Expert - 3 Silver, 5 Bronze; Top Rank 465 world wide
- Hackathon Winner – HackMEE 2.0: Built multilingual e-commerce voice bot in 36 hours.
- Four-time "Top Performer" at Meesho for applied AI innovation.
- Top 1% All-India Rank – Joint Entrance Exam (JEE).